Current-artifact evidence checks in a local molecular workflow: a bounded feasibility report

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Abstract

Scientific workflow agents need evidence that remains valid after input or output files change. We implemented an inspectable local CSV-to-RDKit filtering and report workflow with requirement-to-artifact identities, explicit invalidation and durable execution records. An intended comparison against an equally informed checklist was conditional on native external benchmark readiness. In a bounded examination of eleven metadata-selected ScienceAgentBench tasks, one development reference and six evaluation candidates qualified, but those evaluation candidates covered only three repository lineages against a prespecified minimum of four. The gate therefore failed and no comparative agent matrix or locked generations ran. The primary method effect is not estimable. We report the admission failures, a scorer-interface correction and the resulting software implementation separately from the scientific hypothesis. Actual local-model software acceptance and its associated costs are recorded in the accompanying versioned acceptance table. These author-selected cases do not replace an external benchmark or establish method superiority, independent validation, clinical utility or user benefit.

1. Question and position

The intended question is whether binding an explicit scientific requirement to the identities of the artifacts that support it reduces stale-evidence completion errors compared with an equally informed checklist, under matched model, tool and resource limits. A changed dependency should invalidate the associated status; an unchecked requirement must remain unknown. This is a testable mechanism hypothesis, not a demonstrated contribution in the present report.

ContractBench evaluates temporal validity and byte integrity in tool observations [2]. StateM studies durable execution states and checked transitions [3]. These works make generic state tracking, contracts, hashes and recovery insufficient as standalone novelty claims. Our narrow candidate distinction is dependency-bound completion evidence for mutable scientific files. We have not established that distinction empirically against these or matched controls. Their reported scores are not reproduced here and cannot be transferred to this workflow.

ScienceAgentBench supplies publication-derived scientific programming tasks and execution-based assessment [1]. We attempted a limited native programmatic qualification in the retained verified snapshot, rather than equating a successful model reply with task completion. RDKit supplies the molecular representations, descriptors and drawings used in the delivered tool [4].

2. Methods

2.1 Authority, preservation and resource accounting

The owner authorized a fresh 96 CPU-hour and 24 allocated GPU-hour release, maximum four combined CPU cores and one GPU worker, ending 20 November 2026 at 11:51:07 UTC. Earlier retained 0.675 CPU-hours remain separate. The admission sub-budget was four CPU-hours, one GPU-hour and at most 96 development provider attempts. Conservative reservations are retained after failures; they are not reported as measured CPU consumption. Local guarded work uses one CPU core and an aggregate memory ceiling no greater than the authorized 2 GiB. The owned remote model container uses two CPU equivalents and a single designated GPU. Administrative/control work includes an upfront retained allowance and is not fully individually metered. No paid API, model search or public deployment was used.

Historical ledgers, frozen results and source identities remain preserved. Old final16, private CF1 rows and hidden outcomes were not opened. Rights and source/exposure records accompany the new snapshot; task input, reference and gold/scorer files are separated. Public task snippets and prior exposure are recorded rather than claiming universal contamination freedom. Model pretraining exposure is unknown.

2.2 Native readiness rule

The retained verified archive has SHA-256 `46e715d3b2196d459d2dff52aa487f506a95ec44b44262e82208d086ea879610`. Annotations came from verified revision `9c6e96c9e74572e979b0930ee735041cef528cb7`; the upstream evaluator/environment source was captured at commit `c26e151ed601ba109dc4d35e057ff8e73fec469d`. These identities do not imply reproduction of every paper result or of the entire upstream harness. CodeBERT and paid rubric/image evaluation were not used.

Metadata screening nominated development IDs 2 and 3, and evaluation candidates 5, 16, 18, 19, 22, 26, 40, 41 and 85. The minimum was one qualified development task, six qualified evaluation candidates and four repository-lineage groups among qualified evaluation candidates. Repository lineage is a proxy, not a certificate of publication-level independence. Selection preceded reference-body processing; it is not a random sample of the benchmark. No native comparative matrix was frozen because this prerequisite was not satisfied.

Reference programs and predicates ran in evaluator-only containers. No reference or gold body was fed to the candidate model. Initial mount and byte-transport failures were repaired transparently. A dependency-only repair and one final source-supported compatibility batch were bounded in advance; the final batch ran only the four references that had failed, without refitting successful ones. We stopped after that batch rather than relaxing the cohort rule or searching for a substitute benchmark.

2.3 Scorer correction

The initial adapter expected a scalar result, whereas upstream `eval()` returns `(success, log_info)` [5]. Consequently its earlier nominal false values are unsupported adapter flags, not scientific task failures. We added a strict tuple decoder and rescored retained successful outputs without rerunning their reference fits. Original reports remain present. The final admission receipt carries the corrected interpretation; stale fields retained from the earlier report must be read together with `native_interface_status`. This implementation error belongs to the current work, not to the upstream benchmark.

2.4 Local software and model boundary

The delivered workflow accepts bounded CSV input, computes molecular weight, LogP, TPSA, HBD and HBA using RDKit and applies three inclusive upper limits. Rejected records retain reasons. It exports actual descriptors, selections, reports, evidence and a hash manifest. It does not perform molecular standardization, assay prediction, synthesis or clinical assessment.

Each requirement records supporting artifact hashes and a validator source version. Changes or missing dependencies make that requirement unknown. Current checks cover row accounting, numeric selection consistency and descriptor units; they are not a complete independent scientific oracle. Durable SQLite revisions record attempts, cancellation, failures and explicit resume. Exports are refused after current identity checks fail. The local hash chain is not an external signature or protection against a fully privileged operator replacing every file.

The fixed primary checkpoint is Typhoon 2.1 Gemma3 12B Q4_K_M, using its existing cache and recorded model digest. In product mode it only proposes typed numeric limits and a local export acknowledgement. It cannot upload arbitrary code or choose new tools. Constrained JSON simplifies the task and does not establish general-purpose autonomous scientific reasoning. Every actual provider attempt, raw response, fallback status, generated-token count and resource lease is retained.

2.5 Analysis scope

Author engineering cases were frozen before their product model responses. They cover three threshold requests in English and Thai, with exact parameter and selected-ID checks, plus actual deterministic tool conformance. Each model case has one attempt; failures are retained. No case-directed model tuning or response replacement is allowed. These paired translations are related cases, not six independent scientific samples. We report per-case observations without a population confidence interval, significance test or claims of equivalence.

The intended equal-information checklist, LLM-only, self-debugging, applicable deterministic research controls and invalidation ablation were **NOT_RUN**. No effect estimate or interval can be computed for that unrun comparison. In particular, an absent estimate must not be encoded as a zero effect.

3. Results

3.1 Native admission did not pass

The final receipt qualifies development ID 3 and evaluation IDs 5, 16, 18, 19, 40 and 41. Evaluation candidates represent `psa-lab/predicting-activity-by-machine-learning`, `felixjwong/antibioticsai` and `anikaliu/CAMDA-DILI`: three groups, below four. IDs 2, 22, 26 and 85 remained nonqualifying under the final bounded environment. Missing `mdf_forge` and `fau_colors` and other compatibility failures are retained. This is a bounded environment/reference-readiness result; it neither disproves the proposed mechanism nor demonstrates that ScienceAgentBench is invalid.

Figure 1 and the machine-readable admission table report counts and individual reference status. Their numerators are qualified references, not agent successes. No locked candidate agent program was generated and no method ranking was measured.

3.2 Software and actual model acceptance

See the accompanying versioned actual-run acceptance table and software-check receipt. Component checks exercise actual RDKit subprocesses, evidence invalidation, export tampering, revision conflicts, cancellation and recovery. Browser captures come from the served application with real tool output. These developer checks are software evidence, not independent expert review, customer validation or a substitute external research cohort. Model-case failures, if present, remain in the numerator/denominator and are not repaired into success.

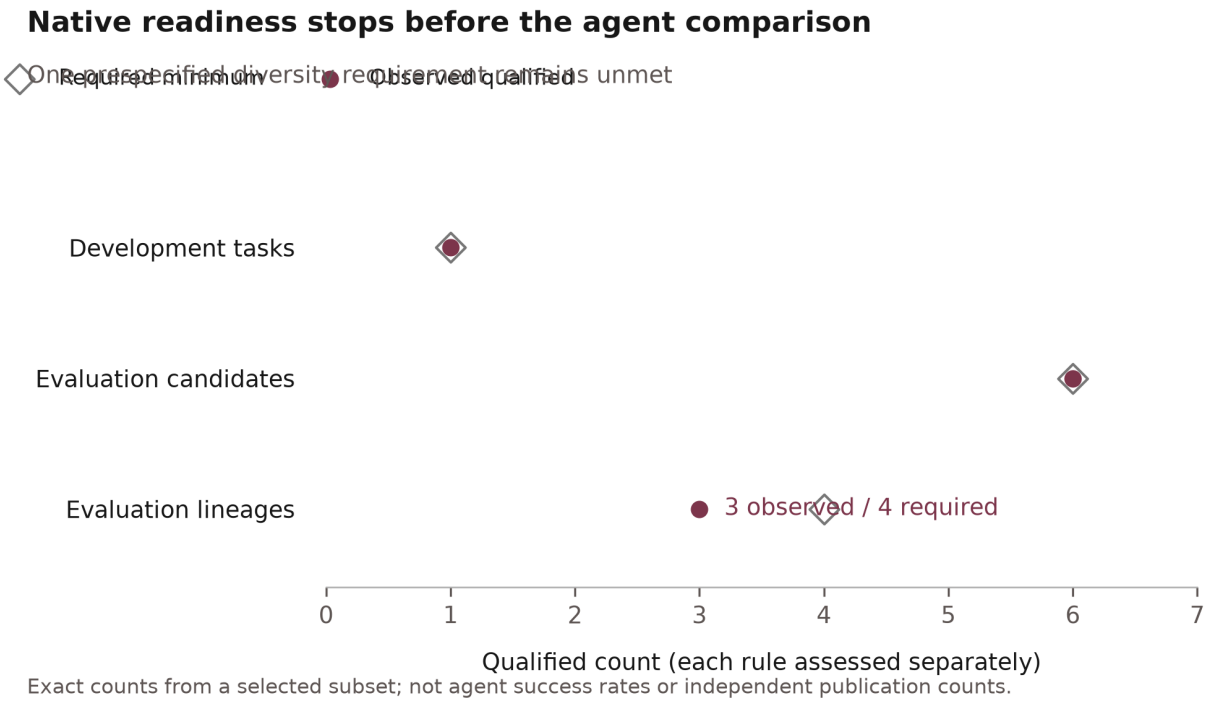
Table 1. Frozen author software cases

Request / language	State	Selected	Calls	Tokens	Wall (s)
MW <=100 / EN	COMPLETED	2	2	56	74.781
MW <=100 / TH	COMPLETED	2	2	56	7.500
LogP <=0 / EN	COMPLETED	2	2	56	7.547
LogP <=0 / TH	COMPLETED	2	2	56	7.859
TPSA <=0 / EN	COMPLETED	1	2	54	8.047
TPSA <=0 / TH	COMPLETED	1	2	54	9.266

All 6/6 cases passed the frozen engineering checks; deterministic conformance passed 3/3. Twelve genuine product responses generated 332 tokens with 3325 prompt tokens. All responses recorded nonzero GPU residency. No case was repeated. These related author-selected cases are not an independent scientific sample.

Timing covers sequential end-to-end job observation, including a cold first model request. No EN/TH or agent/control speed comparison is inferred. Model container observed CPU: 89.452 seconds; actual allocated GPU-container wall: 123.461 seconds. Retained CPU/GPU lease reservations were 2,400/1,200 seconds; local controller work is separate.

Figure 1. The native admission gate did not pass



Qualified evaluation candidates occupy three source-repository lineages against the required four. The threshold remained unchanged. No locked agent program generation or comparative estimate followed.

4. Limitations and interpretation

External readiness failed before the primary experiment, leaving the central research question unanswered. The small metadata-selected subset, adapted wrapper, dependency bounds, source-lineage proxy and known exposure prevent broader benchmark claims. Seven qualified references do not justify lowering a prespecified diversity rule after seeing readiness results. Conversely, the failure does not imply the full benchmark cannot be run with a separately designed environment.

The product case set is small, author-selected and simple; both roles are strongly constrained. Correct JSON, a completed execution and file consistency do not prove the natural-language scientific request was fully answered. A validator can be internally consistent and scientifically wrong. Single-case timing includes hardware and service variability and is not a controlled speed comparison. Research quotas apply to guarded research sessions, not automatically to a user-started server.

No independent evaluator, external user study, wet laboratory, prospective assay, clinical study, business pilot, broad connector ecosystem or demonstrated economic benefit is included. Existing negative research results are retained separately; the new software does not overwrite them or convert them into positive findings.

5. Publication readiness

Not ready for a claim of a superior agent method or a validated pharmaceutical research platform. The honest deliverable is a feasibility/software technical report with reproducible implementation and negative admission evidence. A scientific methods article would need an admitted, independently justified cohort, frozen matched controls, actual model trajectories and uncertainty analyses under a newly reviewed scope. That work is not silently initiated here. A software venue would additionally judge distinctiveness, documentation, sustained software quality and fit; this report does not guarantee acceptance. No manuscript has been sent.

Reproducibility and availability

Source, public aggregate receipts and author-example molecule inputs are in the repository. Raw external benchmark archives, reference/gold bodies, private job stores and full returned environments remain local and excluded from Git. Reacquire third-party data from its official source under the applicable licenses; the code license is not a blanket license for all benchmark content. The new account's README distinguishes public receipts, private exact source snapshots, historical invalid reports and final interpretations. Future reproduction requires the pinned checkpoint and an explicitly configured bounded model backend; the ordinary deterministic local workflow does not need SSH or a model.

References

1. Chen Z et al. ScienceAgentBench: Toward Rigorous Assessment of Language Agents for Data-Driven Scientific Discovery. ICLR 2025; arXiv:2410.05080v3. https://arxiv.org/abs/2410.05080v3 2. Wang J et al. ContractBench: Can LLM Agents Preserve Observation Contracts? arXiv:2605.17281v2, 2026. https://arxiv.org/abs/2605.17281v2 3. Qin Z et al. StateM: Reaching 95.3% Raw Accuracy, or a $15 Frontier Run, on Terminal-Bench 2.1 via Harness Scaling. arXiv:2608.15089, 2026. https://arxiv.org/abs/2608.15089 4. RDKit. Getting Started in Python. Official documentation, accessed 13 November 2026. https://www.rdkit.org/docs/GettingStartedInPython.html 5. OSU-NLP-Group. ScienceAgentBench, `compute_scores.py` and `run_eval.py`, commit c26e151ed601ba109dc4d35e057ff8e73fec469d. https://github.com/OSU-NLP-Group/ScienceAgentBench/tree/c26e151ed601ba109dc4d35e057ff8e73fec469d